

Noncommutative Algebraic Topology I

K-Theory & Homotopical Structures

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1 Introduction

We'll introduce the contents of this text and what you should know before reading it through a list of questions which will be answered. These questions are some of the “usual” ones that any reader would have before reading any text.

1.1 What will you cover?

1.2 What are the prerequisites?

1.3 Where can I go after working through this text?

1.4 What is the historical backdrop of this subject?

1.5 Conventions & General Structure

Most of the structure of the thesis is clear/self-explanatory, so we won't bother getting into the details. There is one major way in which we've re-organized the contents of the thesis that is different from other books/theses and we do want to explain it. Throughout the thesis, you will see boxes of the following form:

1. Definition boxes.
2. Lemma boxes.
3. Proposition boxes.
4. Theorem boxes.
5. Remark boxes.
6. Intuition boxes.
7. Notation boxes.
8. Terminology boxes.
9. Claim boxes.
10. Proof boxes.
11. Motivation boxes.
12. Miscellaneous Results boxes.

Most of these are self-explanatory and we use them whenever we want to note down something that might get potentially referenced again later. This allows us to use the internal referencing feature of LaTeX. The last one requires further explanation. When I was writing this thesis, I intended to go through parts of the literature and fill in details as much as possible. On the other hand, presenting **all** of those details in the main body of the thesis would've been impractical. Moreover, it creates unnecessary difficulties for the reader, particularly those who rely on authors to emphasize what is important and be "quick" about that which isn't so crucial. So, we've successfully found a middle-ground; by placing these "small" (but sometimes, quite technical and difficult) proofs within these miscellaneous results, we create a setting where they can be referenced again if necessary but do not break the narrative structure of the thesis. This should make the thesis much easier to read; one can simply understand the miscellaneous results and move on to the main proofs, perhaps even trying to prove the miscellaneous results themselves.

The best part about the way we've handled this is that these miscellaneous results boxes are directly linked to the corresponding proofs which are located towards the end of the thesis. One simply needs to click the title of a given box (say, **Misc. Res. 4.2.3**) and one is immediately taken to the corresponding miscellaneous proof for that result. Conversely, one can go back to the result from the proof by clicking the title of that proof box. Note that a similar feature does not work for all of the other boxes. This is why we highly recommend having a digital copy of the thesis at hand even if one wishes to use a paper version.

Now, we also have a word or two about standard notation that will be used throughout the thesis. For the most part, there should be no major contradictions between the notation we present here and that which is typically used throughout the thesis. For the sake of clarity, we'll split these up into several boxes that deal with different aspects of the conventions and notation that we'll use. We've chosen these conventions so that they're pretty easy to understand without having to explicitly refer to our choices here. Nevertheless, we have them here for easy reference in the future.

General Mathematical Notation

$\mathbb{K}$ denotes either $\mathbb{R}$ or $\mathbb{C}$.

$\mathbb{F}$ denotes any field.

$\mathbb{N}$ denotes the set of natural numbers (excluding 0)

$\mathbb{N}_0$ is the set of natural numbers including 0

$\mathbb{R}_0^+$ denotes the set of all non-negative real numbers.

$\mathbb{R}^+$ denotes the set of all positive real numbers.

If X, Y are sets, then $\text{Maps}(X, Y)$ is the set of all functions from X to Y .

If A, B are mathematical objects, then $A \stackrel{\text{def}}{=} B$ means that A is defined to be equal to B .

If A, B are statements, then $A \stackrel{\text{def}}{\Leftrightarrow} B$ means that A is defined to be equivalent to B .

Here are some common acronyms we'll use.

Common Acronyms

WLOG - Without loss of generality.

TFAE - The following are equivalent.

TFSH - The following statements hold.

nbhd - neighborhood.

Here are some further general notes for the reader before one begins to read this text.

Notes For The Reader

The reader of these notes should keep the following statements in mind.

1. Any and all mistakes in these notes are my responsibility. If you spot any of these, please contact me immediately at a.vats.math (at) gmail.com^a and let me know so I can fix those mistakes.
2. If you see question marks at any point, then one of the following issues might have occurred:
 - (a) There's something being cross referenced in the text but I haven't used the correct label.
 - (b) There's something in another document which is being cross referenced.
 - (c) There's something that's being cross referenced which doesn't exist.

Whatever the case, do let me know about them because they need to be fixed.

3. If you find a result in this document which does not have a complete proof or anything like that, then it means that I probably just got lazy while writing the proof and haven't gotten back to it yet. I will eventually return to these and fill in the details but it's great if you point them out explicitly.
4. If you find a result in this document for which I've only provided a reference rather than a proof, it's likely because I got quite lazy and postponed writing that proof for a later time. In that case, feel free to inform me about it so I'm forced to actually work through the proof. Moreover, if you have suggestions for improvements to proofs or more conceptual proofs, feel free to reach out as well, you will be appropriately credited.
5. The document makes heavy use of the Cref feature. Of course, you can print whichever bits of it as you want but it's good to have an electronic copy pulled up as well so you may optimize your experience using it.
6. There are certain pdf readers which allow you to hover over a hyperlink in the pdf and get a preview of what's in that hyperlink. This allows you to simply glance at what's in there (which is all that one needs to do sometimes) without having to click the link & actually go to the source. I've listed a couple of pdf readers:
 - (a) For Linux, use GNOME document viewer/Evince.
 - (b) For MacOS, use Skim.
 - (c) For Windows, use Sioyek.

I strongly recommend using these as they vastly improve the experience of reading these notes digitally.

7. The credit for the latex template goes to Senan Sekhon.
8. The credit for the cover page goes to <https://ar.pinterest.com/pin/331999803788929234/>.

If you would like to:

1. contribute to the material in here as an author;
2. comment on the material in here;

then please contact Abhijeet Vats so that further possibilities for collaboration may be discussed.

^aYou may refer to him "Abhi" or "Mousedorff" or any other cool nickname of your choosing.

2 Motivating Projective/Injective Modules

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

3 Constructing Unital Algebraic K_0

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

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| **Motivation.**

4 The Idempotent Picture of Algebraic K_0

Summary

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Conventions

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| **Motivation.**

5 Properties of Algebraic K_0

Summary

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Conventions

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| **Motivation.**

6 Homotopical Structures on Categories of Operator Algebras

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

7 Constructing Operator K_0

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

8 Classifying Unital AF Algebras

Summary

In this lecture, we will discuss the following topics:

1. (Partially) Ordered abelian groups.
2. The order structure of K_0 .

This lecture was written by Abhijeet Vats.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

Let's start with some motivation.

Motivation. In the construction of $\mathbb{Z}$ from $\mathbb{N}$, we made use of the fact that $\mathbb{N}$ comes naturally equipped with a total order to induce a total order on $\mathbb{Z}$. We did this by identifying the “positive cone” of $\mathbb{Z}$, denoted by $\mathbb{Z}^+$, and using that to define the order. Of course, this is the same procedure that one uses for C^* -algebras as well and since $A \in C^*\text{Alg}$ IS equipped with an order, we expect that $K_0(A)$ will also be equipped with a natural order. Our expectations are justified under certain assumptions and the goal of this lecture is to demonstrate that operator $K_0(A)$ does have a nice order structure when A is a unital, **stably finite** C^* -algebra.

9 Axiomatic (Co)Homology I: The Bott Periodicity Theorem

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

10 Axiomatic (Co)Homology II: (Co)Limits & The Mayer-Vietoris Theorem

Summary

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Conventions

Let's start with some motivation.

| **Motivation.**

11 Constructing Operator K_1

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

12 Properties of Operator K-Theory & Some Computations

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

13 Filtration of Operator K-Theory

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

14 The Pimsner-Voiculescu Sequence

Summary

In this lecture, we will cover the following topics:

1. The statement of the Pimsner-Voiculescu Theorem.
2. Two proofs of the Pimsner-Voiculescu Theorem.

Our main goal is to introduce a K-theory comp

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

We will provide motivation in ??, so we will not provide separate motivation here.

15 The Connes-Thom Isomorphism

Summary

In this lecture, we will discuss the following topics:

1. The uniqueness assertion.
2. The existence assertion.

This lecture was written by Abhijeet Vats.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

Let's start with some motivation.

| **Motivation.**

16 Constructing Algebraic K_1

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

17 Homotopical Structures on Rings

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

18 Constructing Negative K-Theory

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

19 Localizations & Lifting Problems

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

Let's start with some motivation.

| **Motivation.**

20 Introducing Model Structures

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

21 A Survey of Examples of Model Categories

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

22 Intrinsic Homotopy in Model Categories

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

23 Comparisons of Model Structures

Summary

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Conventions

Let's start with some motivation.

| **Motivation.**

24 K-Theory: A Categorical Invariant

Motivation

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

Solutions

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